PRACTICAL DBMS

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COURSE CODE: CSA0503

COURSE NAME: DATABASE MANAGEMENT SYSTEMS

Ex. No.: 11 DATABASE WITH AUTO_INCREMENT SEQUENCES

AIM:

To create Auto Increment sequence on the given relation.

DESCRIPTION:

SEQUENCE: In MySQL, a sequence is a list of integers generated in the ascending order i.e., 1,2,3... Many applications need sequences to generate unique numbers mainly for identification e.g., customer ID in CRM, employee numbers in HR, equipment numbers in services management system, etc. To create a sequence in MySQL automatically, you set the AUTO_INCREMENT attribute to a column, which typically is a primary key column.

SYNTAX:

CREATE TABLE table_name(col_name1AUTO_INCREMENT PRIMARYKEY, col_name2, col_name3, ...);

Questions:

1. Populate register number using auto increment in DBMS_Stud table.

2. Manually populate register number

3. Drop the auto increment.

```
mysql> INSERT INTO DBMS_Stud (name, department, age)
    -> VALUES ('Arun', 'CSE', 19);
Query OK, 1 row affected (0.08 sec)

mysql> INSERT INTO DBMS_Stud (name, department, age)
    -> VALUES ('Priya', 'AIDS', 20);
Query OK, 1 row affected (0.05 sec)

mysql> INSERT INTO DBMS_Stud (name, department, age)
    -> VALUES ('Rahul', 'ECE', 19);
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO DBMS_Stud (name, department, age)
    -> VALUES ('Sneha', 'CSE', 20);
Query OK, 1 row affected (0.05 sec)

mysql> SELECT * FROM DBMS_Stud;
+--------+-------+------------+------+
| reg_no | name  | department | age  |
+--------+-------+------------+------+
|      1 | Arun  | CSE        |   19 |
|      2 | Priya | AIDS       |   20 |
|      3 | Rahul | ECE        |   19 |
|      4 | Sneha | CSE        |   20 |
+--------+-------+------------+------+
4 rows in set (0.05 sec)
```

```
mysql> INSERT INTO DBMS_Stud (reg_no, name, department, age)
    -> VALUES (10, 'Karthik', 'MECH', 20);
Query OK, 1 row affected (0.05 sec)

mysql> SELECT * FROM DBMS_Stud;
+--------+---------+------------+------+
| reg_no | name    | department | age  |
+--------+---------+------------+------+
|      1 | Arun    | CSE        |   19 |
|      2 | Priya   | AIDS       |   20 |
|      3 | Rahul   | ECE        |   19 |
|      4 | Sneha   | CSE        |   20 |
|     10 | Karthik | MECH       |   20 |
+--------+---------+------------+------+
5 rows in set (0.00 sec)
```

```
mysql> INSERT INTO DBMS_Stud (name, department, age)
    -> VALUES ('Divya', 'IT', 19);
Query OK, 1 row affected (0.01 sec)

mysql> SELECT * FROM DBMS_Stud;
+--------+---------+------------+------+
| reg_no | name    | department | age  |
+--------+---------+------------+------+
|      1 | Arun    | CSE        |   19 |
|      2 | Priya   | AIDS       |   20 |
|      3 | Rahul   | ECE        |   19 |
|      4 | Sneha   | CSE        |   20 |
|     10 | Karthik | MECH       |   20 |
|     11 | Divya   | IT         |   19 |
+--------+---------+------------+------+
6 rows in set (0.00 sec)
```

```
mysql> ALTER TABLE DBMS_Stud
    -> MODIFY reg_no INT NOT NULL;
Query OK, 6 rows affected (0.09 sec)
Records: 6  Duplicates: 0  Warnings: 0

mysql> DESC DBMS_Stud;
+------------+-------------+------+-----+---------+-------+
| Field      | Type        | Null | Key | Default | Extra |
+------------+-------------+------+-----+---------+-------+
| reg_no     | int         | NO   | PRI | NULL    |       |
| name       | varchar(50) | YES  |     | NULL    |       |
| department | varchar(50) | YES  |     | NULL    |       |
| age        | int         | YES  |     | NULL    |       |
+------------+-------------+------+-----+---------+-------+
4 rows in set (0.09 sec)
```

```
mysql> INSERT INTO DBMS_Stud (reg_no, name, department, age)
    -> VALUES (12, 'Vijay', 'CSE', 20);
Query OK, 1 row affected (0.05 sec)

mysql> SELECT * FROM DBMS_Stud;
+--------+---------+------------+------+
| reg_no | name    | department | age  |
+--------+---------+------------+------+
|      1 | Arun    | CSE        |   19 |
|      2 | Priya   | AIDS       |   20 |
|      3 | Rahul   | ECE        |   19 |
|      4 | Sneha   | CSE        |   20 |
|     10 | Karthik | MECH       |   20 |
|     11 | Divya   | IT         |   19 |
|     12 | Vijay   | CSE        |   20 |
+--------+---------+------------+------+
7 rows in set (0.00 sec)
```

RESULT:

The records from the tables are displayed using auto_increment sequence on the given relation.